



ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ, ಬೆಳಗಾವಿ
VISVESVARAYA TECHNOLOGICAL UNIVERSITY - BELAGAVI

PROPOSED PROJECT SYNOPSIS

by

Sandeep G T

4AL23CS138

Supriya S Pujari

4AL23CS167

Sushma S Shetty

4AL23CS169

Dhruva I S

4AL24CS403

Department of Computer Science and Engineering
Alva's Institute of Engineering & Technology, Moodbidri

Under the Guidance of

Mrs.Deeksha M

Associate Professor, Department of Computer Science and Engineering

Student Signature

Guide Signature

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TITLE OF THE PROJECT

IoT Based Elderly Health Monitoring System

ABSTRACT

The IoT Based Elderly Health Monitoring System is designed to provide continuous health monitoring and safety support for elderly people living independently. The system uses IoT sensors to measure vital parameters such as heart rate, body temperature, SpO₂ level, and motion activity. It consists of a wearable unit for health monitoring and fall detection, and a room monitoring unit for activity tracking. The collected data is sent to the cloud using ESP32/NodeMCU, and alerts are automatically sent to caregivers if abnormal conditions like falls, abnormal vital signs, or inactivity are detected. This project aims to provide a reliable, low-cost, and real-time smart healthcare solution for improving elderly safety.

INTRODUCTION

Elderly people often suffer from health problems such as heart disease, high blood pressure, diabetes, and mobility issues. Continuous monitoring is required to ensure their safety, but regular hospital visits are difficult and expensive.

With the advancement of IoT technology, remote health monitoring has become possible. Sensors can collect real-time health data and transmit it to caregivers through the internet. This improves emergency response time and reduces health risks.

Traditional monitoring systems mainly depend on manual panic buttons, which may fail if the elderly person becomes unconscious. Therefore, an automated monitoring system is required.

This project proposes an IoT based elderly monitoring system that combines health monitoring, fall detection, and activity monitoring to provide a complete safety solution.

PROBLEM STATEMENT

Elderly individuals living alone face serious risks due to delayed medical assistance during emergencies such as falls, heart attacks, or unconsciousness. Existing monitoring systems either depend on manual interaction or generate high false alarms.

Problem:

To develop an IoT based elderly monitoring system that continuously monitors health parameters, detects falls, analyzes activity patterns, and provides emergency alerts with reduced false alarms.

LITERATURE SURVEY

Several research works have been done in the area of elderly healthcare monitoring:

1. Mubashir et al. (2013) studied fall detection techniques using wearable sensors.
2. Patel et al. (2012) discussed wearable health monitoring systems.
3. IoT healthcare systems using ESP32 provide real-time monitoring.
4. AI based monitoring systems help in detecting abnormal behavior.
5. Cloud based healthcare platforms enable remote patient monitoring.

These studies show that combining IoT sensors with AI techniques can improve elderly healthcare monitoring.

OBJECTIVES OF THE PROPOSED PROJECT

The main objectives are:

- To design a wearable device to monitor heart rate and oxygen level
- To detect falls using accelerometer sensors
- To monitor room activity using motion sensors
- To send real-time data to cloud platform
- To generate alerts during emergency conditions
- To reduce false alarms using multi-parameter verification

PROPOSED METHODOLOGY

Step-by-Step Process of the Proposed System

1. Data Collection:

The wearable sensors (MAX30102 and MPU6050) collect vital health parameters like heart rate, SpO₂ level, and fall movement. At the same time, the room unit sensors (PIR and DHT11) collect motion and environmental data.

2. Data Processing:

The ESP32 (wearable unit) and NodeMCU (room unit) process the collected sensor data and check whether the values are within normal limits.

3. Data Transmission:

The processed data is transmitted to the IoT cloud platform using WiFi modules for storage and monitoring.

4. Condition Analysis:

The system analyzes the data to detect abnormal conditions such as fall detection, abnormal heart rate, low oxygen level, or long inactivity.

5. Alert Generation:

If any abnormal condition is detected, the system sends emergency alerts to caregivers or family members through a mobile app, SMS, or notification.

6. Data Display:

The health data is displayed on a mobile dashboard or cloud platform for continuous remote monitoring.

EXPECTED OUTCOME OF THE PROPOSED PROJECT

- Continuous health monitoring
- Fall detection system
- Real-time alert system
- Remote monitoring dashboard
- Reduced emergency response time
- Improved elderly safety

SUMMARY

The IoT Based Elderly Health Monitoring System provides a smart solution for monitoring elderly health and safety. By combining wearable sensors and environmental monitoring, the system ensures continuous monitoring.

The system improves emergency response by detecting abnormal conditions automatically and notifying caregivers. This project helps elderly people live independently while ensuring their safety.

In future, this system can be enhanced by adding AI prediction, ECG monitoring, and mobile healthcare applications.

REFERENCES (IEEE FORMAT)

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